

(1) For the x vs. y data to the right:

(a) use the least squares method for a linear equation, viz., $y = ax + b$ using perpendicular offsets to determine the slope and intercept. Explain how you determined the correct root of the quadratic to use for the slope. Also, calculate the quality of the fit Q^2 .

(b) fit the same data to a straight line using vertical offsets and calculate the Q^2 for this fit.

(c) and finally, repeat part (b) for horizontal offsets.

Use the definitions from lecture to work this problem.

x	y
1	4.5
2	11
3	8
4	13.9
5	12
6	14
7	18
8	25
9	19
10	23.6

(2) (a) Find two different ways to linearize the equation $r = Az/(B + z)$ where A and B are adjustable constants. The reaction rate is r , and the concentration is z . Use both linearizations to fit the data to the right using perpendicular offsets. Compare the values of the constants for the two linearizations and the quality Q^2 of the fits.

(b) Use the solver on Excel to find a numerical solution for fitting the nonlinear equation to the data using vertical offsets. You will need a reasonable first guess for the solver in order to get the numerical method to converge. Use one of your results in part (a) for this guess. Compare this result with the two results in part (a).

z	r
2.50E+00	20
4.00E+00	43
1.00E+01	65
2.00E+01	106
4.00E+01	112
1.00E+02	128
2.00E+02	150
1.00E+03	165

(3) Murphy 1.2 Solve using Cramer's rule.

(4) Murphy 1.15

(5) Murphy 1.17

65/84.

x	y	(xi-Sx)^2	(yi-Sy)^2	(xi-Sx)(yi-Sy)
1	4.5	20.25	108.16	46.8
2	11	12.25	15.21	13.65
3	8	6.25	47.61	17.25
4	13.9	2.25	1	1.5
5	12	0.25	8.41	1.45
6	14	0.25	0.81	-0.45
7	18	2.25	9.61	4.65
8	25	6.25	102.01	25.25
9	19	12.25	16.81	14.35
10	23.6	20.25	75.69	39.15

Sx = 5.5
 Sy = 14.9
 Sxx = 30.25
 Syy = 222.01

average(x)
 average(y)

SSxx = 8.25
 Ssyy = 38.532
 Ssxy = 16.36
 B = -0.925489

$(xi - Sx)^2$
 $(yi - Sy)^2$
 $(xi - Sx)(yi - Sy)$
 $(SSxx - SSyy)/2SSxy$

it is negative because

a(perp) = 2.28803
 b(perp) = 2.31581
 m^2(perp) = 0

$-B - B\sqrt{1 + \frac{1}{B^2}}$
 $Sy - a_{\perp} Sx$

Lambda1 = 45.68224225
 Lambda2 = 1.099757751
 Q^2(perp) = 0.908177975

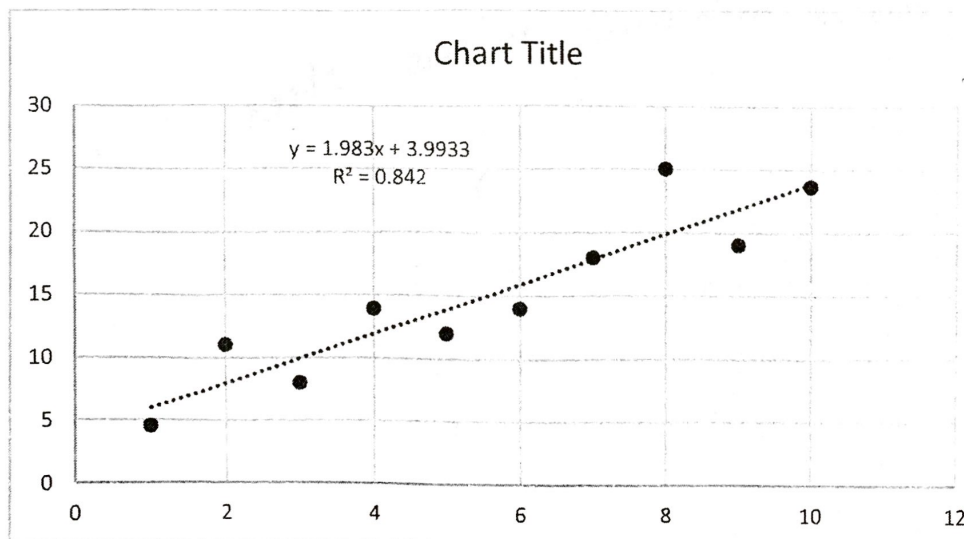
$\rightarrow \frac{1}{2} [SSxx + SSyy + \sqrt{(SSxx - SSyy)^2 + 4(SSxy)^2}]$
 $\rightarrow \frac{1}{2} [SSxx + SSyy - \sqrt{(SSxx - SSyy)^2 + 4(SSxy)^2}]$
 $\left(\frac{\lambda_1 \sqrt{m^2 + 1} - \sqrt{\lambda_1^2 m^2 + \lambda_2^2}}{\lambda_1 \sqrt{m^2 + 1} + \sqrt{\lambda_1^2 m^2 + \lambda_2^2}} \right)^2$

a(vert) = 1.98303
 b(vert) = 3.99333
 m^2(vert) = 0.00303

Q^2(vert) = 0.786330064

a(hori) = 2.35526
 b(hori) = 1.94609
 m^2(hori) = 0.00011

Q^2(hori) = 0.900225341



x

y

1/x

1/y

(xi-Sx)^2

(yi-Sy)^2

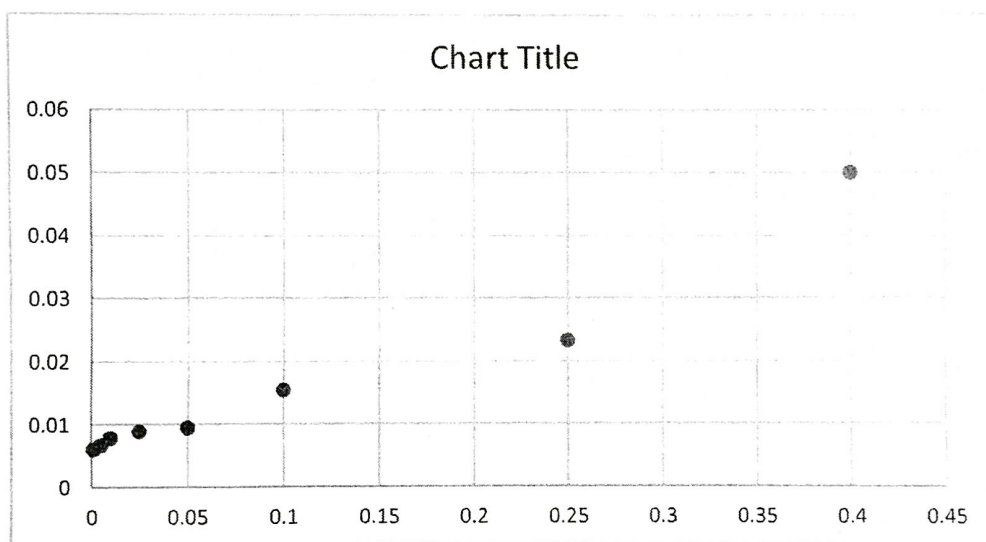
(xi-Sx)(yi-Sy)

2.5	20	0.4	0.05	0.0869513	0.00115989	0.010042604
4	43	0.25	0.02325581	0.0209888	5.34796E-05	0.001059467
10	65	0.1	0.01538462	2.627E-05	3.11617E-07	2.86091E-06
20	106	0.05	0.00943396	0.0030388	4.23655E-05	0.000358802
40	112	0.025	0.00892857	0.00642	4.92E-05	0.000562018
100	128	0.01	0.0078125	0.0090488	6.61025E-05	0.000773399
200	150	0.005	0.00666667	0.010025	8.60474E-05	0.000928777
1000	165	0.001	0.00606061	0.010842	9.76586E-05	0.001028988

Sx, Sy, SSxx, Ssyy, Ssxy =

 $1/y = B/(Ax) + 1/A$

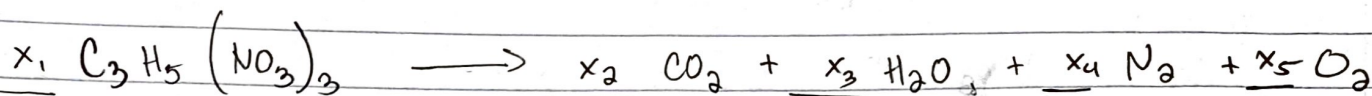
0.105125	0.01594284	0.0184176	0.000194382	0.001844615
B =	4.93957603	lambda1 =	0.018602452	
a(perp) =	0.10020684	lambda2 =	9.53891E-06	
b(perp) =	0.0054086			
m^2 =	0	Q^2 =	0.997950995	



a. → needs another linearization method

b7-6

#1.2



C	3	1	0	0	0
H	5	0	2	0	0
N	3	0	0	2	0
O	9	2	2	0	2

matrix A

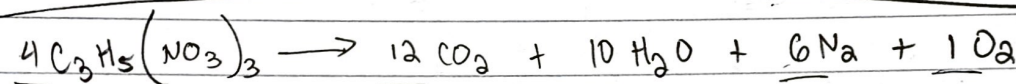
$\rightarrow \text{ref}(\text{matrix A}) \rightarrow$

1	0	0	0	4	= x ₁
0	1	0	0	-12	= x ₂
0	0	1	0	-10	= x ₃
0	0	0	1	-6	= x ₄

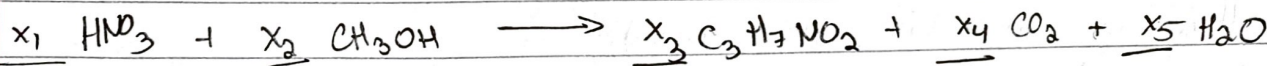
free variable, could be anything I want

$$x_5 = 1$$

product side so it's negative



#1.15



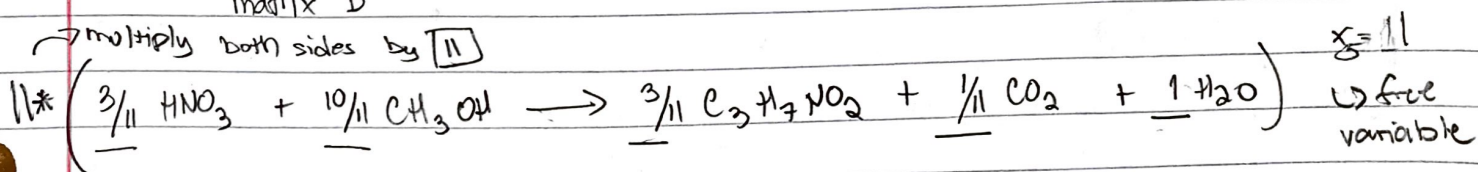
H	1	4	7	0	2
N	1	0	1	0	0
O	3	1	2	2	1
C	0	1	3	1	0

matrix B

$\rightarrow \text{ref}(\text{matrix B}) \rightarrow$

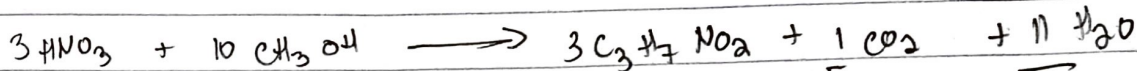
1	0	0	0	3/11	= x ₁
0	1	0	0	10/11	= x ₂
0	0	1	0	-3/11	= x ₃
0	0	0	1	-1/11	= x ₄

multiply both sides by 11



$$x_5 = 11$$

free variable



part b.?

1.17

$$C = 12.0107 \text{ u}$$

$$H = 1.00784 \text{ u}$$

$$O = 15.999 \text{ u}$$

Total mass of compound

$$1) C_2H_5OH = 2(12.0107) + 6(1.00784) + 15.999 = 46.06744 \text{ u}$$

$$2) CH_4 = 12.0107 + 4(1.00784) = 16.04206 \text{ u}$$

$$3) H_2O = 2(1.00784) + 15.999 = 18.01468 \text{ u}$$

$$4) C_6H_{12}O_6 = 6(12.0107) + 12(1.00784) + 6(15.999) = 180.15228 \text{ u}$$

$$Wt\% = \frac{\text{total mass of H}}{\text{total mass of compound}} \times 100\%$$

$$1) \frac{6(1.00784)}{46.06744} \times 100\% = \underline{13.13\%}$$

$$2) \frac{4(1.00784)}{16.04206} \times 100\% = \underline{25.13\%}$$

$$3) \frac{12(1.00784)}{18.01468} \times 100\% = 11.19\%$$

$$4) \frac{12(1.00784)}{180.15228} \times 100\% = \underline{6.713\%}$$

Some of the claims are absurdly unrealistic like the 50%. The molecule should be composed of high number of hydrogen first before having 50% of the total weight of the compound. At the end of the day the weight % of hydrogen still depends on the amount of hydrogen in the compound and the amount of other elements in it.